

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (20 points)*Submit the postlab to the TA at the end of the lab.***Measuring Sound Speed in Air (8 points)****Question 3****1 point**

Record the frequency of the tuning forks and the distances to the first, second, and third nodes in Data Table 1.

Data table 1: *Node lengths in the resonant air column, driven by a tuning fork.*

Frequency f (Hz)	Node Distance L_1 (m)	Node Distance L_2 (m)	Node Distance L_3 (m)

Question 4**2 points**

Calculate the average distance between consecutive nodes ($L_{i+1} - L_i$) for each frequency. Recall that in a standing wave, adjacent nodes are separated by exactly half a wavelength, so this distance equals $\lambda/2$. Show a sample calculation below.

Question 5**2 points**

Calculate the speed of sound in air for each of the three tuning forks. Show your work.

Question 6**2 points**

From your three measurements of the sound speed, compute the average velocity of sound in air and its uncertainty using the standard deviation of your values. Show your work.

Question 7**1 point**

Using your average sound speed and its uncertainty from the previous question, compare your result with the value predicted by the following formula for the speed of sound in dry air as a function of temperature:

$$v(T) \approx v_0 + \left(0.6 \frac{\text{m/s}}{^\circ\text{C}}\right) T, \quad v_0 = 331.5 \text{ m/s at } 0^\circ\text{C},$$

where T is the air temperature in degrees Celsius. (Note: room temperature is about 20°C).

Sonometer: Standing Waves in a Wire of Known Density (2 points)**Question 8****1 point**

Using eq. (1) and eq. (3), compute the resonant frequency of the fundamental harmonic. Show your work.

Question 9**1 point**

Record the number of antinodes and the resonant frequency (calculated and measured) for a given length of string L , a given tension T , and a given linear density μ in Data Table 2.

$L =$ _____ m $T =$ _____ N $\mu =$ _____ kg/m

Data table 2: *Measurements of the wire of known density.*

Number of antinodes	Calculated resonant frequency (Hz)	Measured period τ (s)	Uncertainty in measured period $\Delta\tau$ (s)	Measured resonant frequency (Hz)

Sonometer: Standing Waves in a Wire of Unknown Density (10 points)**Question 10****1 point**

Record the color code of the unknown wire below. In Data Table 3, record the tension, the node spacing, and the period of the wave.

Data table 3: *Measurements of the wire of unknown density.*

Tension T (N)	Node spacing (cm)	Period τ (s)	Uncertainty in period $\Delta\tau$ (s)

Question 11**3 points**

Calculate the velocity and the uncertainty in the velocity of the wave for each of the five tensions using the measured wavelength and the period of the fundamental harmonic. (Hint: how do you relate the period of the fundamental harmonic as measured on the oscilloscope to its frequency?) Propagate the uncertainty in the period $\Delta\tau$ to the velocity using

$$\Delta v = \frac{\lambda \Delta\tau}{\tau^2},$$

which assumes the wavelength has no uncertainty. Compute the uncertainty in the square of the velocity as

$$\Delta(v^2) = 2v \cdot \Delta v.$$

Add your results to Data Table 4, showing work for one calculation out of the five.

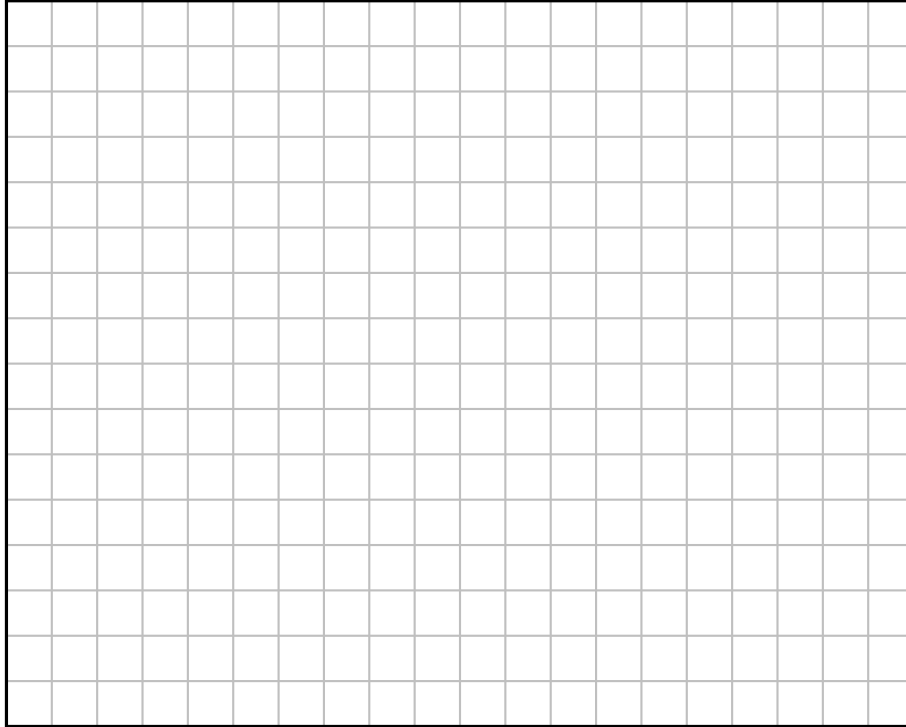
Data table 4: *Calculations of the wire of unknown density.*

Tension T (N)	λ (m)	f (Hz)	v (m/s)	Δv (m/s)	$\Delta(v^2)$ (m ² /s ²)

Question 12**3 points**

Plot v^2 versus T in Graph 1. Include labels, units, and the uncertainty on v^2 for each point.

Graph 1: *Plot of v^2 vs. T .*



Is the relationship linear? Find the slope of the line. The slope is $1/\mu$, where μ is the linear density of the unknown wire. Show your work.

Question 13**2 points**

Calculate the uncertainty in $1/\mu$ as follows. In Graph 1, draw the minimum and maximum slopes $m_{\min}$ and $m_{\max}$ that are consistent with the upper and lower error bars of the data. Use these slopes to compute an average slope and an uncertainty on the average:

$$\overline{m} = \frac{|m_{\max} + m_{\min}|}{2},$$
$$\Delta\overline{m} = \frac{|m_{\max} - m_{\min}|}{2}.$$

Question 14**1 point**

How does the value of μ from the slope compare to the accepted value of the color-coded wire? (Ask your TA for the accepted value of μ for your unknown wire.) Do the values agree within experimental uncertainties?